



Using Context-Guided data Augmentation, lightweight CNN, and proximity detection techniques to improve site safety monitoring under occlusion conditions

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ABSTRACT

Automated recognition of image patterns in surveillance cameras is beneficial for safety monitoring. A representative application is visual proximity detection for accident prevention. However, existing deep learning (DL)-based proximity detection methods are likely to be inaccurate due to congestion, background clutter, and occlusion in construction sites. Another problem is that most of these methods cannot suffice computing capability, network training efficiency, and performance accuracy at the same time. This study first presents a novel context-guided data augmentation method that addresses the performance loss issue due to object occlusion. This method places new instances on images based on contextual image information rather than randomly erasing images to simulate occlusion circumstances. Next, a light-weight and less-costly DL model named YOLOv4-EFNB0 is formulated for multi-class construction resource detection in real-time. Moreover, this study presents a proactive proximity detection application by integrating YOLOv4-EFNB0 with homography transformation. Based on the experiment, YOLOv4-EFNB0 demonstrates a better object detection outcome when trained on the augmented dataset over the Moving Objects in Construction Sites (MOCS) baseline dataset. This study also conducts a feasibility test for the visual proximity detection application. The results indicate the application has a high accuracy rate and a fast speed, namely, 96.76% in 25 Frames Per Second (FPS), which can facilitate the automation of safety monitoring and inspection.

1. Introduction

Construction is a large and complex industry that provides numerous job opportunities all over the world (Govender et al., 2019; Hou et al., 2017). Given the dynamic, ever-changing, and diverse spatiotemporal nature of the industry environment, it is perceived as a high-risk sector. In the United States, fatalities in the construction industry reached 1008 in 2020, which are higher than the other industries (LaborStatistics, 2021). In Australia, more than half (59 %) of the worker fatalities were associated with vehicle collisions (41 %), falls from heights (11 %), and trapped-by accidents (7 %) (SafeWorkAustralia, 2021). There have been fruitful research works around ameliorating the workforce environment through developing different safety precautionary methods (Hou et al., 2020; Luo et al., 2022; Zhang et al., 2021). These works have also proved that most onsite accidents could have been prevented if consistent safety monitoring process was established, and real-time safety monitoring can play a role (Awolusi et al., 2019; Cheng et al., 2022; Hou et al., 2021).

Safety monitoring requires the collection of information on construction activities and the identification of hazardous circumstances. A conventional method for safety monitoring is through regular inspection and patrol conducted by safety inspectors, through which preventative and corrective actions can be put in place. It is identified that traditional safety management methods have limitations because they are mainly relied on human judgement, which could be subjective and biased (Paneru and Jeelani, 2021). During construction, it is also unlikely to identify and act on imminent incidents timely and accurately. Nowadays, a viable solution would be utilising automation technology, as evidenced by a vast amount of research works ranging across unsafe behaviour monitoring (Fang et al., 2020b), construction activity recognition (Slaton et al., 2020), unauthorised intrusion monitoring (Jin et al., 2020), and so on.

An important research topic for automated safety monitoring is onsite proximity detection because worker and machinery location tracking can alert struck-by accidents. While past research showed the

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potential of Radio Frequency Identification, Ultra-Wideband, and Bluetooth Low Energy technologies (Awolusi et al., 2019), a real-world construction site would not allow vast deployed sensors which bring in interruption and other safety concerns. Cameras are low-cost and have been widely used in construction sites. Logistics contexts can be acquired and analysed based on the camera-generated video to derive movement patterns and infer unsafe scenarios. However, this information retrieval and analytics is typically manual-based, time-consuming, and lack of efficiency. Fortunately, cutting-edge technologies such as computer vision and Deep Learning (DL) have been developed to enhance the efficacy of surveillance cameras. Computer vision enables automatic pattern recognition, and data-driven DL introduces end-to-end learning where image features can be extracted without hand-crafting engineering (Luo et al., 2018; Xu et al., 2021). Integrating DL and computer vision is a promising solution to the ineffectiveness of surveillance cameras since it can proactively identify hazards and predict unsafe behaviour from video analytics. Different types of DL models have been investigated. Kong et al. (2021) leveraged a Long Short-Term Memory (LSTM) model and a SiamMask model to predict worker movement trajectory and alert workers entering a hazardous area. Kolar et al. (2018) used Visual Geometry Group-16 as a DL model backbone to identify safety guardrails. Fang et al. (2018c) utilised two Convolutional Neural Network (CNN) models to detect if workers are wearing safety harnesses while working at heights. Xiao and Kang (2021b) employed a You Only Look Once (YOLO)-based model to track multiple construction machines for construction safety monitoring.

Despite the evolvement of contemporary vision-based DL models in construction safety monitoring, object occlusion in proximity detection seems to be a long-standing issue that has not been well tackled (Saleh et al., 2021; Xiao et al., 2020; Zhao et al., 2019). The image-forming principle has determined that the depth of field for spatially occluded objects may not be correctly derived from a single camera, therefore, detection errors for occluded objects are very common, for instance, when a worker stands in front of an excavator captured by a camera. According to Wang et al. (2020a), a viable solution is to train the DL model using the image datasets that contain rich occlusion features, which is also known as the ‘minimising the partial occlusion effect’ method. Unfortunately, it is revealed from our review that there are inadequate image databases that contain rich entity occlusion features for DL model training (Li and Li, 2022; Son et al., 2019). Even if this problem is solved, deriving the ground truth of occlusion features, namely, annotating all the occluded entities and letting the DL algorithm understand ‘what is what’, could be another problem (this process is laborious and time-consuming). In answer to this problem, Bang et al. (2020) attempted to generate the artificial occlusion features by overlaying certain instances onto images. A downside of this, however, is that the DL model may mistakenly identify the non-occluded objects as occluded ones in the real-world onsite environment, which is also known as overfitting (Naveed, 2021; Shorten and Khoshgoftaar, 2019). Last but not least, how to improve the DL model processing speed for proximity detection, how to increase the accuracy of distance measurement, and how to interpret the depth information from raw images are all the remaining issues that worth investigation (Paneru and Jee-lani, 2021). The relevant works have been identified from our review as well. To illustrate, Kim et al. (2016) introduced vision-based fuzzy inference algorithms into proximity detection, so that the detection accuracy can be slightly enhanced; Kim et al. (2019) leveraged a super-computer to compensate the DL model performance loss. According to Paneru and Jeelani (2021); Thompson et al. (2020), these works did not address the detection accuracy for entities of non-linear motions, and were lack of practicality for resource-constrained construction scenarios either.

Therefore, this study proposes two research questions, which are: 1) how to enhance the quality of DL training datasets for occluded entity identification, and 2) how to process real-time proximity detection using low computational power. The remainder of this paper is organised as

follows: Section 2 reviews the current DL-based applications in construction safety and the related works on dataset development and augmentation, Section 3 describes the methodologies in detail, Section 4 presents the experimental results and discusses the system practicality, and Section 5 concludes the study and future work.

2. Literature review

Research on vision-based construction safety management has investigated how to identify risk hazards from imagery data. Most previous works utilised vision-based object detection methods to recognise various entities in construction sites, such as workers, construction machinery, protective equipment, and so on. This review is three-facet, namely, the different vision-based applications for construction safety, the object detection algorithm development state-of-the-art, and the existing image datasets and data augmentation methods, followed by a summary of the certain research gaps.

2.1. Automatic monitoring applications for construction safety management

With the help of DL algorithms, image and video-based monitoring of construction sites has the potential to improve construction workforce safety. Failure to wear Personal Protective Equipment (PPE) is one of the most common unsafe risks (Fang et al., 2020a). Fang et al. (2018a) developed a CNN-based method to detect the PPE use of steplejacks. The proposed method achieved high precision in ever-changing environments. Nath et al. (2020) investigated the performance of three DL-based approaches for PPE recognition, with the pros and cons revealed. Fang et al. (2018b) introduced a Faster Region-based CNN (Faster R-CNN) model to accurately identify hardhats from far-field surveillance videos. Li et al. (2022) utilised YOLOv5 to conduct PPE standardised use inspection by considering the spatial relationship between workers and PPE. Cheng et al. (2022) combined a worker re-identification technique and PPE classification to monitor safety compliance among workers across multiple cameras. Ding et al. (2018) enhanced the accuracy of unsafe action recognition by coupling CNN and LSTM. As construction work is physically demanding, inappropriate working postures of workers may cause injuries. Thus, automated postural ergonomic assessment plays an essential role in construction safety. Zhang et al. (2018) estimated 3-Dimensional (3D) skeletons and joints of workers from video sequences using a multi-stage CNN model. Non-ergonomic postures can be recognised accurately in this study. Yu et al. (2019b) presented an ergonomic assessment tool by integrating CNN and two LSTM layers. This study was the first to introduce a 2-Dimensional (2D) video-based 3D pose estimation onto ergonomic assessment. Yu et al. (2019a) developed a CNN-based physical fatigue assessment by computing joint torques based on 3D skeletons. These studies have demonstrated the effectiveness of DL-based applications for construction safety monitoring from the perspective of object identification and classification.

Many unsafe situations in construction also relate to proximity between workers and heavy machinery. DL models can detect objects and extract their pixel coordinates in an image (Yang et al., 2013). Based on the pixel coordinates, the pixel distance between objects can be calculated. Several image post-processing techniques are used to convert pixel distance to actual distance (Sutton et al., 2008). The derived actual distance is essential for collision prevention. Brilakis et al. (2011) employed a couple of cameras to obtain two 2D local coordinates of a construction entity. The 3D global coordinate of the entity can be derived by using triangulation. Luo et al. (2020) proposed a monitoring system that analyses construction workers and their movement patterns by 2D image post-processing techniques. They applied the safety-checking rules that specify the distance thresholds to the system. Each piece of construction equipment had a predefined warning level according to the threshold. By doing this, the system automatically popped

a warning when someone entered hazard zones. Kim et al. (2017) and Wu et al. (2022) integrated AR into proximity detection to help onsite workers view orientational and locational information about construction equipment. With this integration, struck-by and fall-from-height risks can be detected. According to their results, however, the processing speed was low to 5 Frames Per Second (FPS) and the average latency was high to 280 ms. Because real-time safety monitoring has a high requirement on detection latency, the previous development did not add substantial value to the safety practice. Therefore, optimising the DL model performance through algorithmic and parametric adjustment is essential.

2.2. DL-based object detectors

CNNs are a well-known type of DL models that have been commonly used for object detection (Kulchandani and Dangarwala, 2015). CNNs with multiple convolutional layers have the capability of extracting and self-learning image features. With this characteristic, a well-trained CNN can classify and segment image patterns autonomously. Faster R-CNN and Mask R-CNN are typical CNN-based object detectors that have high localisation and recognition accuracy. They first generate certain Regions of Interest (ROIs) that have a high probability of containing an object from the image. Then, the last layer of the CNN predicts the outcome based on the ROIs. These detectors are mainly applied to structural health monitoring (Nayyeri et al., 2019), such as road damage detection (Wang et al., 2018) and deterioration classification (Fujita et al., 2020; Hasuike et al., 2022). However, recent research has suggested that R-CNNs may not be suitable for monitoring construction dynamics due to their complex computing pipelines (Kumar et al., 2022).

In recent years, single-stage CNN models (e.g., Single Shot Detector (SSD) and YOLO) have prevailed. Unlike R-CNNs, they consider object detection as a simple regression problem by predicting bounding boxes and computing class probabilities at the same time, which can significantly reduce the computing complexity and the processing time. Roberts and Golparvar-Fard (2019) formulated a one-stage CNN (RetinaNet) to monitor the operational status of excavators and dump trucks; Jung et al. (2021) developed a one-stage 3D CNN for the real-time tracking of construction machinery movement with a speed of 36.6 FPS; Wu et al. (2019) employed YOLOv3-tiny to detect workers and their PPE, which achieved 31.3 FPS and 63.4 % Average Precision (mAP); Nath et al. (2020) further improved the mAP of YOLOv3 to 74 %. Despite the considerable results, previous research mainly adopted high-performance Graphics Processing Units (GPUs) to realise the high processing speed and performance, and such GPUs are still too expensive to vast deploy. Even though there are light-weight versions for YOLOv3 and YOLOv4, their accuracy is inadequate, namely, 22 % mAP when tested on the MS COCO dataset (Bochkovskiy et al., 2020). These limitations stymie the use of those models. A trade-off option could be optimising CNN model structures. This solution can be referred to from Guo et al. (2021), which compared different activation functions and loss functions in YOLOv4 to find optimal performance in railroad component inspection. However, research on optimising onsite object detectors is relatively scant. Therefore, how to refine the model structure is a critical topic for onsite proximity detection.

2.3. Image dataset development and augmentation for DL

The efficacy of DL models in object detection can benefit from the expansion of imagery datasets (Russakovsky et al., 2015). At present, there are numerous image datasets available to the public (Table 1). Everingham et al. (2010) developed the Pascal VOC dataset, which is widely used as a benchmark in the DL community. This dataset contains 9,963 images and 24,640 annotated objects under 20 object categories. The MS COCO dataset is another popular dataset that contains 2,500,000 labelled instances from 328,000 images (Lin et al., 2014). In

Table 1

Public image datasets dedicated to object detection.

Related articles	The size of dataset	The number of object categories	Include construction resources	Occluded entities are labelled sufficiently
Everingham et al., (2010)	9,963 images	20	NO	NO
Lin et al., (2014)	328,000 images	80	NO	NO
Tajeen and Zhu (2014)	2,000 images	5	YES	NO
Roberts and Golparvar-Fard (2019)	500 videos	2	YES	NO
Wu et al. (2019)	3,174 images	1	YES	NO
Xuehui et al. (2021)	42,000 images	13	YES	NO

the construction context, Tajeen and Zhu (2014) established an image dataset for measuring the performance of construction machine recognition models. However, this dataset was not suitable for contemporary DL model training as it contains less than 2,000 labelled images that covers only 5 machinery types. This databased was further enriched by Xiao and Kang (2021a). Roberts and Golparvar-Fard (2019) published a video-based dataset that contains approximately 500 construction activity videos of excavators and trucks. Wu et al. (2019) published 3,174 images that range across a variety of safety helmet use cases from different construction sites, and annotated 18,893 safety helmet instances using bounding boxes. Xuehui et al. (2021) formulated a database named Moving Objects in Construction Sites (MOCS) and it is comprised of 42,000 onsite images and 13 construction object classes. However, it is worth noting that none of these state-of-the-art datasets contains a sufficient amount of ground truth labelled occlusion entities that are ground truth labelled for object detection.

Data augmentation is an essential technique to improve the DL model robustness. Conventional image data augmentation methods include translating, scaling, rotating, image mirroring, colour changing, and random cropping (Ding et al., 2016). In recent years, advanced image augmentation techniques such as automatic data augmentation, regional-level augmentation (e.g., Cutout/Random Erasing), and image-level augmentation (e.g., AutoAugment/Fast Augmentation) are emerging (Hataya et al., 2020; Jiang et al., 2020; Walawalkar et al., 2020). Charalambous and Bharath (2016) proposed a simulation-based methodology to generate synthetic videos for training DL-based gait recognition models. Kolar et al. (2018) utilised Autodesk Revit to generate 3D guardrail models. Then, synthetic guardrails are pasted onto construction images. By doing this, a large number of construction images with synthetic guardrails can be obtained. Bang et al. (2020) introduced a cut-and-paste method by placing various construction equipment onto images. However, a common limitation is that these methods do not consider the actual contextual information of the images, resulting in irrational image semantics (e.g., trucks in the sky). As well, the DL model trained from these image datasets may incur the overfitting problem and erroneous recognition (Dvornik et al., 2018). A likely solution would be to place object instances at certain locations in a picture based on the surrounding context. But having said that, it is necessary to formulate explicit context-guided augmentation that can address occlusion conditions.

3. Methodology

This study aims to develop a low-cost proximity detection system that can efficiently track the position and proximity of onsite entities. First, a novel data augmentation method is proposed for coping with occlusion conditions. The proposed method can generate synthetic

images through overlaying various occlusion patterns onto contextual image clues. Next, a light-weight DL model is developed to meet the low computational requirement without compromising the model training efficacy. This ensures the model can be readily applied to construction scenarios. Last but not least, the DL model is integrated with optimised image post-processing techniques to provide real-time proximity information between workers and construction entities in surveillance videos.

3.1. Development of the context-guided data augmentation method

Common data augmentation methods involve, but are not limited to, image rotation, cropping, flipping, and colour space conversion. Although these methods can significantly upsize the data, the following training process can be quite time-consuming. Compared with these common data augmentation methods, a novel method known as CutMix that splices two images to synthesize seems to be more robust (Yun et al., 2019), as it allows partially viewed objects to be correctly recognised by the DL model in one training epoch. Bochkovskiy et al. (2020) formulated the YOLOv4-based mosaic data augmentation method which further improves the CutMix efficacy through synthesising multiple images into one epoch. However, this image stitching technique has limitations as it does not examine the image context or allow the object detector to learn intraclass variability of targets.

In answer to the limitations of the state-of-the-art data augmentation methods, this paper proposes a new method named context-guided data augmentation to generate context-aware occluded images. We first develop a context-guided model using ResNet (He et al., 2016), which can quickly locate the construction resources from an image and generate a bounding box containing the detected construction resource and its surrounding regions. After that, the context-guided model randomly places a segmented construction-related instance onto the bounding box. This way the construction resource in the bounding box can be effectively occluded in different proportions. The augmented examples developed by existing methods and our proposed method are shown in Fig. 1, followed by an elaboration on the image augmentation rationale.

3.1.1. Selection of candidate regions for context-guided image stitching

The rationale of this approach is to utilise the CNN-based context model to locate existing image objects and generate candidate regions to showcase synthetic objects. To effectively undertake multi-class image identification tasks, the popular ResNet50 network pretrained on

ImageNet was employed as the context model (He et al., 2016). Once we input the image to the context model, the candidate regions can be automatically defined as a set of bounding boxes. Given directly stitching instances to these bounding boxes may lead to occluded areas on the detector, it is therefore suggested a simple heuristic be formulated to generate larger bounding boxes that cover the single object surroundings.

3.1.2. Stitching instances onto candidate regions

Construction-related instances are stitched onto the corresponding location whenever the previous procedure generates a candidate region. Linear blur is selected to reduce the noise and artefacts during this procedure.

3.1.3. Image annotation for the augmented data

Once the image is enhanced by stitching new objects, the annotations are modified accordingly. We use LabelMe (Russell et al., 2008) to draw the tightest box around a new object and assign its class to the box. The output of LabelMe is saved as the JSON format and the YOLO training model labels are saved as the TXT format. Hence, the formatted files with labelled information can be converted to TXT files accordingly.

3.2. Development of the light-weight DL model

3.2.1. YOLOv4 architecture

YOLOv4 was introduced by Bochkovskiy et al. (2020) and is an advanced DL model nowadays. The overall performance of YOLOv4 on the MS COCO dataset is impressive. The mAP is increased significantly compared to the previous version without dropping the FPS. YOLOv4 has three significant upgrades. First, CSPNet (Wang et al., 2020b) modifies Darknet53 to CSPDarknet53, which fulfills low-level feature fusion and robust feature extraction. As shown in Fig. 2(b), the feature map of the base layer is split into two sections. This splitting and merging strategy allows for more feature fusion at different levels. Second, Spatial Pyramid Pooling (SPP) (He et al., 2015) is used (Fig. 3) to introduce three different pooling processes on the final output to generate fixed-length feature maps, which allows the input image size to be adjusted. Third, Feature Pyramid Networks (FPN) (Lin et al., 2017) are replaced by Path Aggregation Network (PANet) (Liu et al., 2018) (Fig. 4(b)).

3.2.2. EfficientNet

EfficientNet has three significant advantages. First, the network is

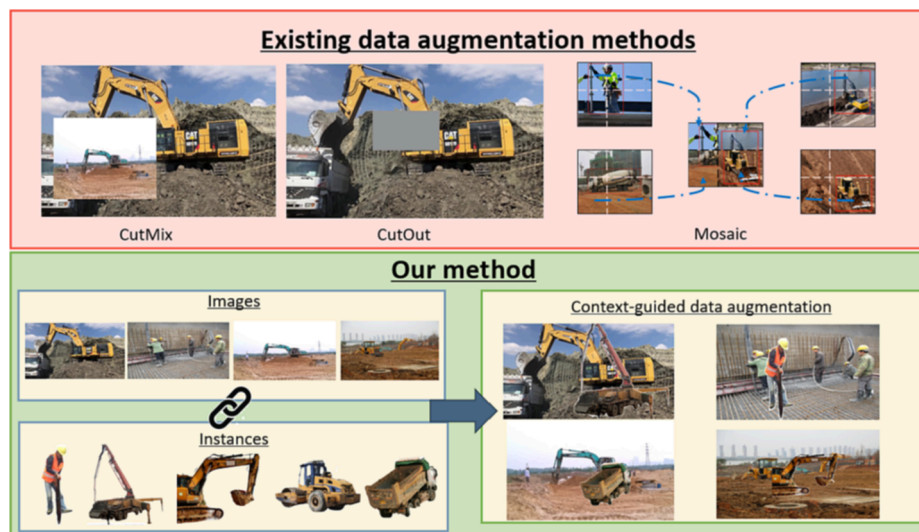


Fig. 1. Augmented examples using existing methods and our proposed method.

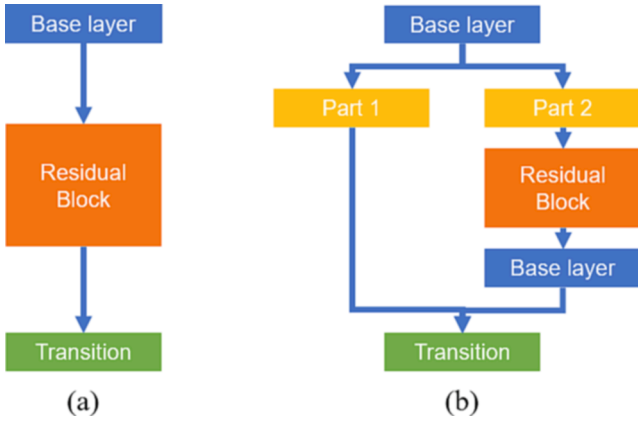


Fig. 2. Comparison between (a) Darknet53 and (b) CSPDarknet53.

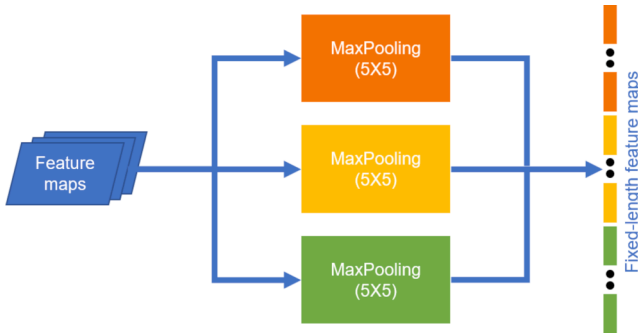


Fig. 3. Spatial pyramid pooling.

deepened. Feature extraction through deeper layers is well achieved by using skip connection. Second, more features are obtained by increasing the number of convolutional layers. Third, the network can learn and express more features by enhancing the input resolution. EfficientNet scales the network width, depth, and resolution with a compound coefficient Φ , and maximises the network accuracy on resource-constrained computing. The relationship between the compound coefficient and the network characteristics is shown as follows:

$$\text{Depth} : d = \alpha^\Phi \quad (1)$$

$$\text{Width} : w = \beta^\Phi \quad (2)$$

$$\text{Resolution} : r = \gamma^\Phi \quad (3)$$

$$\alpha \geq 1, \beta \geq 1, \gamma \geq 1 \quad (4)$$

where α , β , and γ are the resource allocations for the network depth,

width, and resolution. According to (Tan and Le, 2019), the optimal coefficients of the network are: $\alpha = 1.2$, $\beta = 1.1$, and $\gamma = 1.15$. EfficientNet comprises a stem module and seven blocks. The attention mechanism and residual structure are used to develop the blocks. There are eight versions of EfficientNet, which are from B0 to B7. B0 is most compact and computationally-efficient to feature extraction. In this study, EfficientNet-B0 (Fig. 5) is used to replace the backbone network of YOLOv4.

3.2.3. YOLOv4-EfficientNet-B0

Although the original YOLOv4 model can perform well in object detection, its parameters and weights are relatively extensive. Therefore, the backbone network CSPDarknet53 was replaced by EfficientNet-B0 in YOLOv4. Block 3, Block5 and Block7 were selected as the effective feature extraction layers. There is a size discrepancy between the outputs of the three feature extraction layers of CSPDarknet53 and those of EfficientNet-B0. Connecting the output of Block 3, Block 5, and Block 7 through 2D convolution layers allows us to change the feature size such that it matches the output size and allows for additional feature extraction. The rest of the structure is the same as the original YOLOv4 model. Fig. 6 demonstrates the architecture of YOLOv4-EFNB0 (YOLOv4-EfficientNet-B0).

The loss function of YOLOv4-EFNB0, comprised of confidence loss, classification loss, and regression model loss, is equivalent to that of YOLOv4. It is worth noting that compared to YOLOv3, the mean squared error is replaced by the Complete-Intersection over Union (CIoU) (Zheng et al., 2020) to produce better outcomes in regression loss. The loss function of CIoU is expressed as follows:

$$LOSS_{CIoU} = 1 - IoU + \frac{\rho^2(b, b^{gt})}{c^2} + \alpha v \quad (5)$$

$$d = \frac{\rho^2(b, b^{gt})}{c^2} + \alpha v \quad (6)$$

where $\rho^2(b, b^{gt})$ stands for the distance between the centroid of the ground truth and the centroid of the prediction bounding box, and c stands for the diagonal distance of the area that covers both the bounding box and the ground truth. Fig. 7 demonstrates the simplified CIoU model.

The formulas of α and v are described as follows:

$$\alpha = \frac{v}{1 - IoU + v} \quad (7)$$

$$v = \frac{4}{\pi^2} \left(\arctan \frac{w^{gt}}{h^{gt}} - \arctan \frac{w}{h} \right)^2 \quad (8)$$

3.3. Development of the proximity detection application

The proximity detection application integrates YOLOv4-EFNB0 and

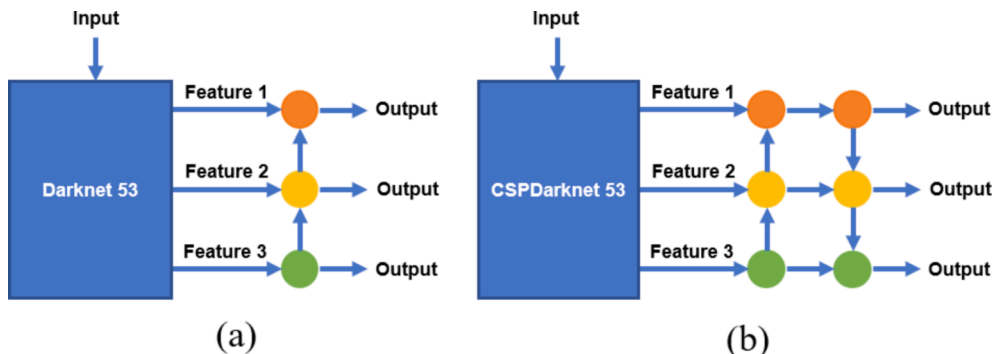


Fig. 4. Comparison between (a) FPN and (b) PANet.

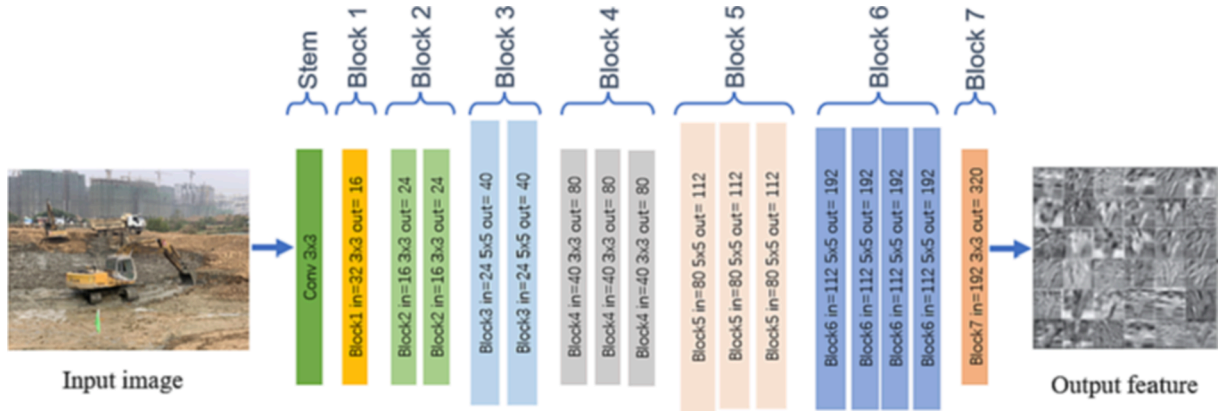


Fig. 5. The architecture of EfficientNet-B0.

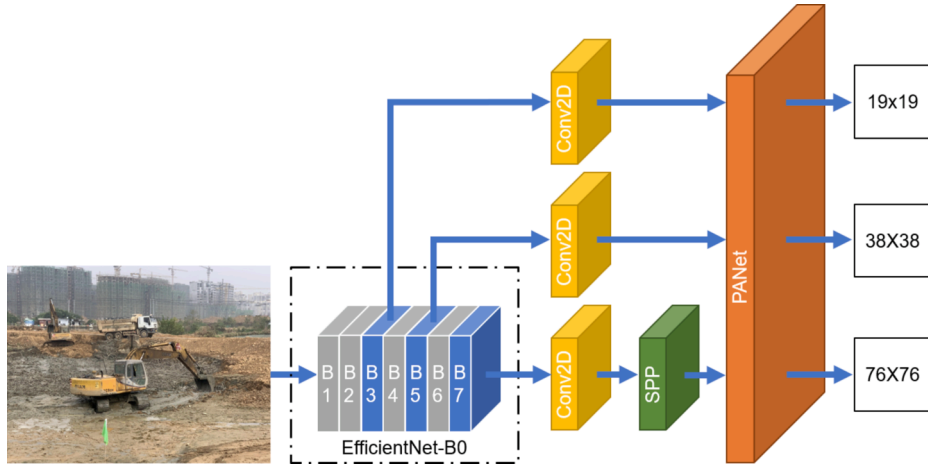


Fig. 6. The network structure of YOLOv4-EfficientNet-B0.

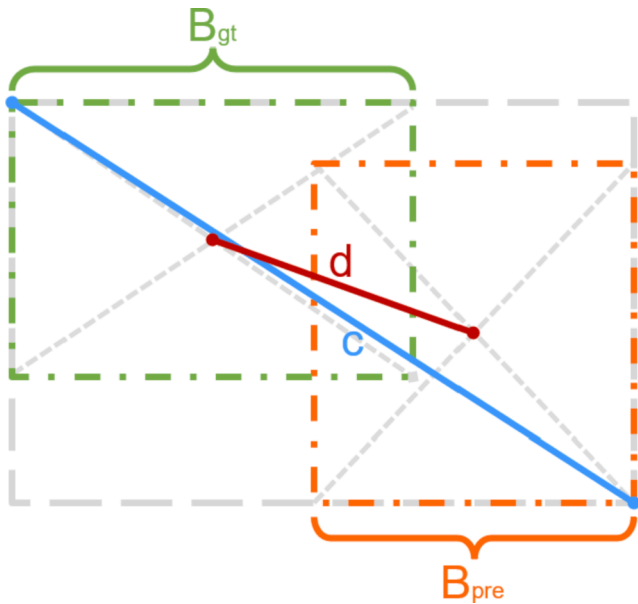


Fig. 7. The simplified model of CloU.

the homography transformation to identify workers and excavators using video frames and then estimates the distance between the centroids of the bounding boxes. When the distance limit is not maintained,

the target bounding box will become red and the screen will show the total number of violations. The workflow of the proximity detection application is shown in Fig. 8.

Video footage collected by surveillance cameras provides an angle view of a construction site. Calculating the distance of each centroid of bounding boxes would lead to significant displacement errors. To improve the accuracy of the proximity measurement, a bird's view of the construction site is needed. This can be derived by implementing a homography transformation to the coordinates of the selected four points from the original camera view. These four points can be derived as shown below:

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & 1 \end{bmatrix} \begin{bmatrix} k \\ t \\ 1 \end{bmatrix} \quad (9)$$

where k and t are the pixel coordinates of one of the four points in the camera view. x and y are the coordinates of the same point in the transformed view. M_{ij} ($i = 1, 2, 3$; $j = 1, 2, 3$) is the coefficient of the homography matrix, which the direct linear transformation can calculate.

After finishing the bird's view transformation, the pixel distance of a worker and equipment in the image can be transformed to metric distance by using the Euclidean equation:

$$Euclidean distance = W * \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \quad (10)$$

where W is a scale factor derived by a reference object's actual size

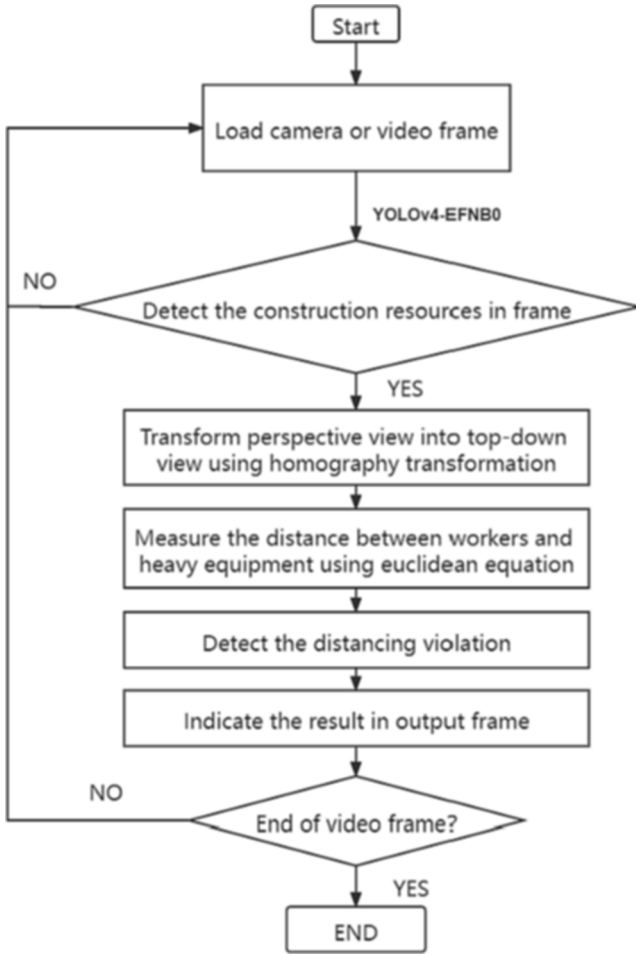


Fig. 8. The workflow of the proximity detection application.

divided by its pixel size. (x_1, y_1) and (x_2, y_2) are the pixel coordinates of the worker and the excavator, respectively.

A variety of conditions are set to decide the bounding box colour after the measurement is finished. The three conditions, namely, Number of workers $(W) \geq 1$, Number of excavators $(E) \geq 1$, Distance between (W) and (E) is ≤ 10 m, are defined to judge whether or not a worker in a frame is too near to an excavator. The model would return false truth results if incorrect detection occurs. All the data are collected to validate the accuracy of the application.

4. Experiments

In this section, three preliminary experiments have been conducted to validate the effectiveness of the data augmentation method, compare the performance of the construction resource detector against other cutting-edge DL models, and test the feasibility of proximity detection in an actual construction video.

4.1. Baseline dataset

The MOCS (Xue et al., 2021; Xuehui et al., 2021) is our baseline dataset. The dataset contains approximately 40,000 images, including 13 types of construction entities. To evaluate whether the proposed augmentation method is effective in a small dataset, we downsized the baseline data to 9 types of construction resources (6,141 images) in our experiments, including workers, mobile cranes, drum rollers, bulldozers, excavators, concrete mixer trucks, concrete pump trucks, tower cranes, and dump trucks. These resources are common in various earthwork

activities, such as digging and lifting (crane and excavator), hauling and loading (dump truck, bulldozer, and concrete mixer truck), and compacting and finishing (concrete pump truck, worker, and drum roller). Following the procedures described in Section 3.1, we generated 2,534 augmented images into the baseline dataset and named them MOCS-DA. The class distribution in both datasets is shown in Fig. 9. Then, YOLOv4-EFNB0 will be trained on both datasets separately.

4.2. Implementation details

The experiment in this paper was run under the PyTorch Object Detection API, which was published by the Facebook AI team. The PyTorch version is 1.8.2, and the Python version is 3.7. The experimental environment was a 64-bit Win10 Pro, and the CPU was an i5-7500 with frequencies of 3.40 GHz. An NVIDIA GeForce GTX 1080Ti GPU was used for acceleration. The cuDNN (version 7.6.5) and CUDA (version 11.3) packages were used to promote the training process. In terms of the general training setting, the total training epoch was 100, the batch size was 16, and a total of 14,685 iterations were performed during the training. For the first 20 epochs, the learning rate was set to 0.001 to accelerate the model convergence. For the following epochs, the learning rate was changed to 0.0001 to increase the accuracy of the model. Table 2 demonstrates the basic configuration of the local computer. Fig. 10 showcases the loss convergence during the training process. The learning curves for both sets demonstrate that the augmented dataset provides more valuable feature information for the models to learn.

4.3. Evaluation metrics

To evaluate the detection performance, PASCAL VOC metrics have been applied for object detection and classification. The Intersection over Union (IoU) can be calculated by the area of overlap between a predicted box (B) and a ground truth bounding box (B_{gt}) divided by the area of their union:

$$IoU = \frac{\text{Area}(B \cap B_{gt})}{\text{Area}(B \cup B_{gt})} \quad (11)$$

A detection result is considered a True-Positive (TP) only when it meets three requirements: the confidence score is larger than 0.7; the predicted category matches the category of the ground truth; and the $IoU \geq 0.5$. If the result fails to meet the last two criteria, it is considered a False-Positive (FP). Moreover, the PASCAL VOC metrics include additional assumptions to determine TPs and FPs. If the ground truth is associated with multiple bounding boxes, only the one with the highest confidence score serves as a TP. When the confidence score of detection is lower than the confidence threshold, the result serves as a False-Negative (FN). As per these rules, the formulas for precision, recall, and F1-score are as follows:

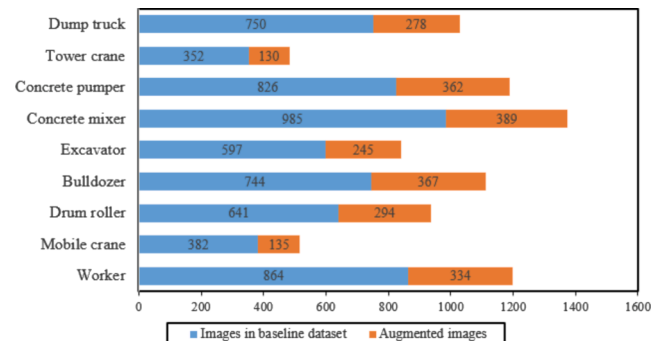
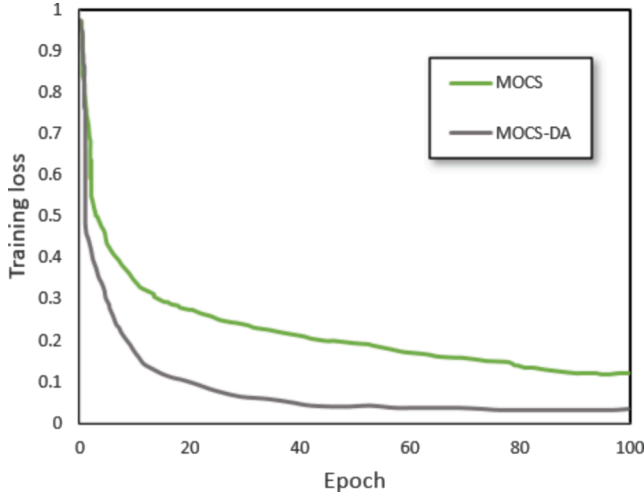


Fig. 9. The class distribution in both baseline dataset and augmented images.

Table 2

The basic configuration of the local computer.

Computer configuration	Parameters
Operating system	64-bit Win10 Pro
GPU	NVIDIA GeForce GTX 1080Ti GPU
Central Processing Unit (CPU)	i5-7500
Random Access Memory (RAM)	16G

**Fig. 10.** Training loss curve of YOLOv4-EFNB0 trained on both datasets.

$$Precision = \frac{TP}{TP + FP} \quad (12)$$

$$Recall = \frac{TP}{TP + FN} \quad (13)$$

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} \quad (14)$$

Precision measures the ratio of TPs to the sum of TPs and FPs. Recall is the ratio of TPs to the total number of target objects. The F1 score is determined as the harmonic mean of precision and recall. To evaluate the overall performance of the developed DL model, Average Precision (AP) was applied. AP is measured by the area under the Precision-Recall curve of a certain class. As YOLOv4-EFNB0 is developed for multi-object detection, the mAP measured by the mean of the AP values of all categories is also used as an evaluation metric:

$$mAP = \sum_{i=1}^k \frac{AP_i}{k} \quad (15)$$

5. Results and discussion

In this section, the experimental results of YOLOv4-EFNB0 have been analysed in depth, and a comparative analysis has been conducted to illustrate the strengths of the data augmentation method and the proposed DL model. The proximity detection application has been tested on a real-world construction video to validate its feasibility. Moreover, the practical implications of this study have been discussed.

5.1. Experimental results of construction resource detection with/without using an augmented dataset

To illustrate the effectiveness of the proposed augmentation method on a construction-related dataset, YOLOv4-EFNB0 was trained on MOCS and MOCS-DA separately with the same weight initialisation and training settings. The ratios of the training set, validation set, and test set are 7:1:2. The test results are shown in Table 3. It is evident that the data

augmentation method improves the general model performance. However, the AP values of some categories, such as mobile and tower cranes, are not ideal. Due to the shape of cranes, the pixels that display the crane are only a small number of the pixels contained in the bounding box. This means that the crane can be largely occluded, or even completely occluded, by newly placed object instances. Therefore, in this case, the augmented data hinder the DL model from learning the features of cranes, resulting in a decrease in the AP values. As shown in Fig. 11, some representative examples are generated by the two methods. It is noted that YOLOv4-EFNB0 trained on MOCS-DA achieves better performance in occluded object detection and small object detection. Specifically, as shown in Fig. 11(c) and Fig. 11(i), YOLOv4-EFNB0 trained on MOCS-DA can detect the truck behind the excavator correctly, whereas the one trained on MOCS fails to detect the occluded truck and trucks that are far from the camera. As shown in Fig. 11(b), the detector incorrectly recognises the uniform behind rebars as workers. The main reason is that the features of uniforms and workers are too similar. Additionally, the image with occlusion may induce the detector to be too sensitive to noise, resulting in incorrect detections. As shown in Fig. 11(e) and Fig. 11(k), although other machines do not occlude the excavator, the detector trained on MOCS still cannot identify it. The main reason is that the most identifiable part of the excavator, which is the bucket, is missing. In contrast, the detector trained on MOCS-DA can detect the excavator accurately, which showcases the effectiveness of using this augmentation method. Similar findings can be observed in Fig. 11(f) and Fig. 11(l), where the worker sitting on the excavator can be detected by the detector trained on MOCS-DA, whereas the detector trained on the MOCS dataset fails to do it. It is believed that detectors trained on a large-scale dataset with context-guided data augmentation are more robust to different fields of view, illumination levels, and occlusion conditions.

5.2. Comparative analysis of YOLOv4-EFNB0 against other DL models

To evaluate whether the performance of YOLOv4-EFNB0 is compromised due to its light-weight characteristic, YOLOv3, YOLOv4, and Faster R-CNN with ResNet were used to perform the comparative analysis. These three models are all trained on both the MOCS and MOCS-DA datasets with the same weight initialisation and training settings. Fig. 12 shows the Precision-Recall curve of these four models trained on two datasets. It is noted that models trained on MOCS-DA have robust performance with higher recall and precision rates. Table 4 shows the different model sizes, parameter amounts, and Floating-point Operations Per Second (FLOPs). To better visualise the results, several histogram graphs were created (Fig. 13 and Fig. 14). The histogram graph contains five performance values (AP^{small} , AP^{large} , F1 score, mAP, and inference speed). AP^{small} is defined as the AP for an object's pixel smaller than 63×63 , and AP^{large} refers to the AP for an object's pixels larger than 63×63 . It is obvious that the overall

Table 3

Comparison of performance metric results obtained with YOLOv4-EFNB0 model with/without data augmentation.

Categories	YOLOv4-EFNB0			
	Trained on MOCS		Trained on MOCS-DA	
	F1 score	AP	F1 score	AP
worker	0.68	0.42	0.85	0.64
crane	0.64	0.39	0.57	0.36
roller	0.62	0.38	0.70	0.53
bulldozer	0.61	0.47	0.72	0.54
concrete mixer	0.70	0.34	0.88	0.65
pump truck	0.56	0.44	0.63	0.49
tower crane	0.38	0.32	0.33	0.29
dump truck	0.62	0.48	0.71	0.54
excavator	0.70	0.53	0.74	0.56
mAP		0.41		0.52

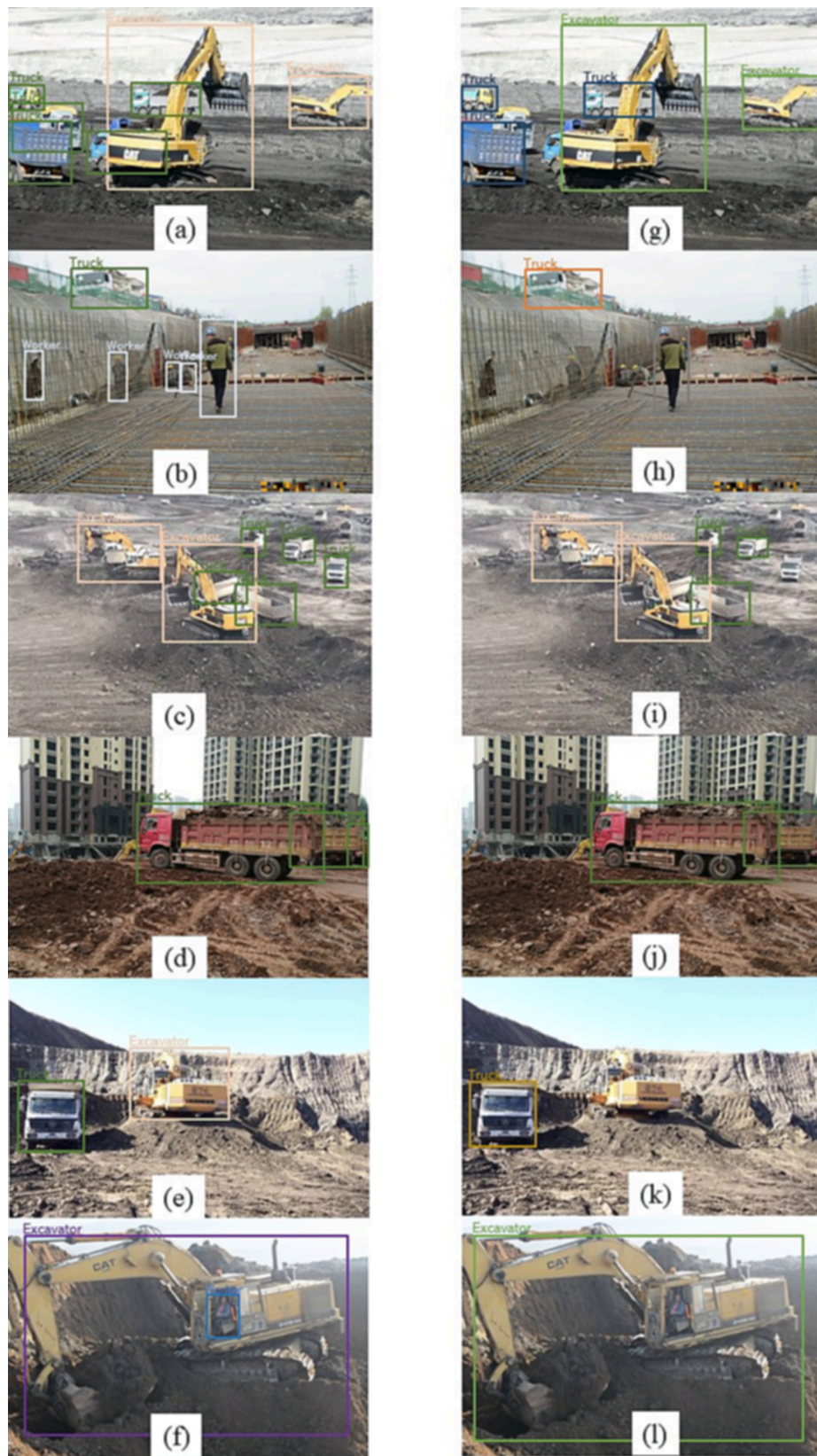


Fig. 11. (a)-(f) Test examples produced by YOLOv4-EFNB0 trained on MOCS-DA; (g)-(l) Test examples produced by YOLOv4-EFNB0 trained on MOCS.

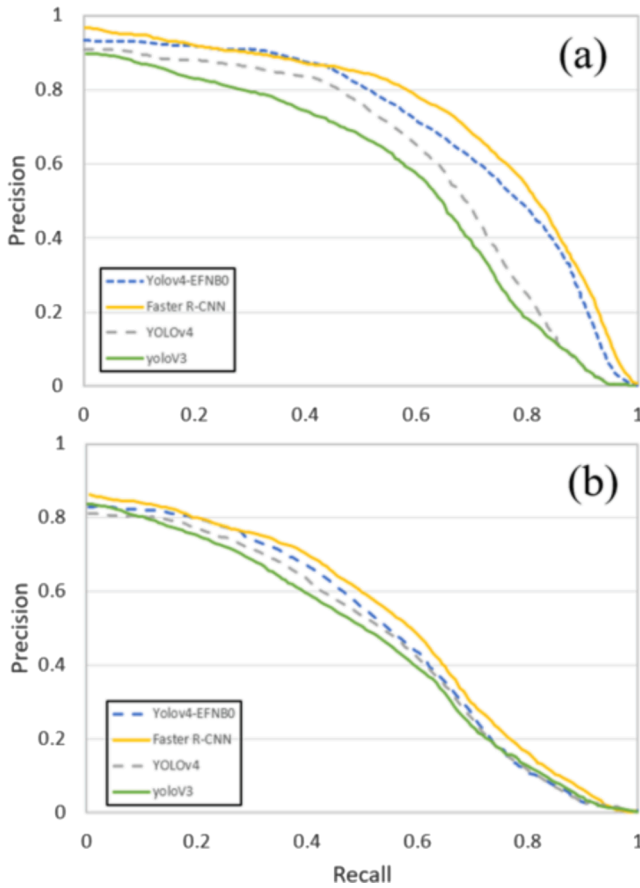


Fig. 12. Precision-recall curves of the testing model: (a) models trained on the MOCS-DA dataset; (b) models trained on the MOCS dataset.

Table 4

Computational cost comparison between different models.

DL models	Size (MB)	Parameter	FLOPs
YOLOv3	235	6.15×10^7	3.21×10^{10}
YOLOv4	244	6.40×10^7	2.98×10^{10}
Faster R-CNN with ResNet	165	4.15×10^7	1.49×10^{11}
YOLOv4-EFNB0	136	3.76×10^7	1.21×10^{10}

performance of YOLOv4-EFNB0 and other detectors trained on the MOCS-DA dataset yielded better performance than those trained on the baseline dataset. Although the AP rates of Faster R-CNN performed

better than that of YOLOv4-EFNB0, the two-stage detector has more calculations (1.49×10^{11}), vastly decreasing the inference speed. YOLOv4-EFNB0 is much better than YOLOv4 in terms of computational cost, where the weight size is decreased to 136 MB, the parameter amount is decreased to 3.76×10^7 , and the calculation amount is reduced to 1.21×10^{10} . Notably, the performance of YOLOv4-EFNB0 trained on the MOCS dataset did not achieve desirable results, and its performance in small object detection was even lower than that of YOLOv4 trained on the MOCS dataset. The main reason may be that the detector cannot learn enough features from the dataset by the light-weight backbone. Although the light-weight characteristic makes YOLOv4-EFNB0 more suitable for practical usage in construction sites, it is essential to apply data augmentation methods to enhance the training efficiency.

According to the results of comparative analysis, the model size of YOLOv4-EFNB0 is tremendously reduced with negligible compromise in accuracy. The overall performance of YOLOv4-EFNB0 trained on the MOCS-DA datasets is better than that of the state-of-the-art DL models.

5.3. Feasibility test for proximity detection at the construction site

To validate the feasibility of the proximity detection application, a case study based on an earthmoving project was conducted. First, YOLOv4-EFNB0 was applied to detect the construction entities in videos. We filtered the detected classes into “worker” and “excavator” in this case study. Once these classes were detected, their positions were converted from the image coordinates into the transformed coordinates. Second, as described in Section 3.4, we selected four points in the video frames to determine the projective view of the construction site. Next, the actual distance between four points in and the corresponding pixel coordinates of these points in the frames were measured accordingly. Then, these four points were used to process the homography transformation. The last step was to conduct the proximity measurement and determine if there was a violation on the construction site. It is noted that only the entities within the projective view were considered (Fig. 15).

This application was evaluated on a video from an earthmoving project containing workers and excavators. The test video contains 4,410 frames, and the ground truth of each frame has been annotated accordingly. The Mean Absolute Error (MAE) (Chai and Draxler, 2014) of the physical distance between the worker and the excavator over all frames was selected to evaluate the proximity detection application performance. The equation of MAE_d is:

$$MAE_d = \frac{1}{n} \sum_{i=1}^n |d_g - d_i^{min}| \quad (16)$$

where d_i^{min} is the closest distance between a particular worker and an

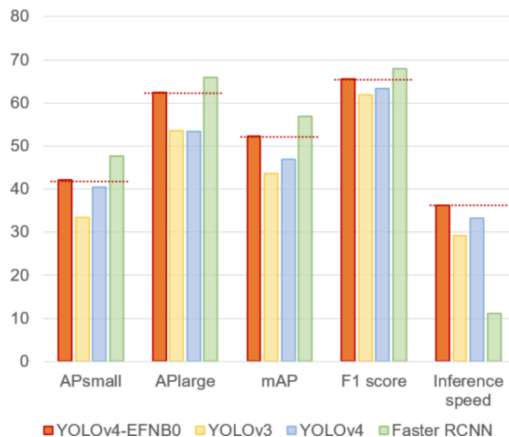


Fig. 13. Comparison of the state-of-the-art models trained on MOCS-DA: YOLOv3, YOLOv4, and Faster R-CNN with ResNet-152.

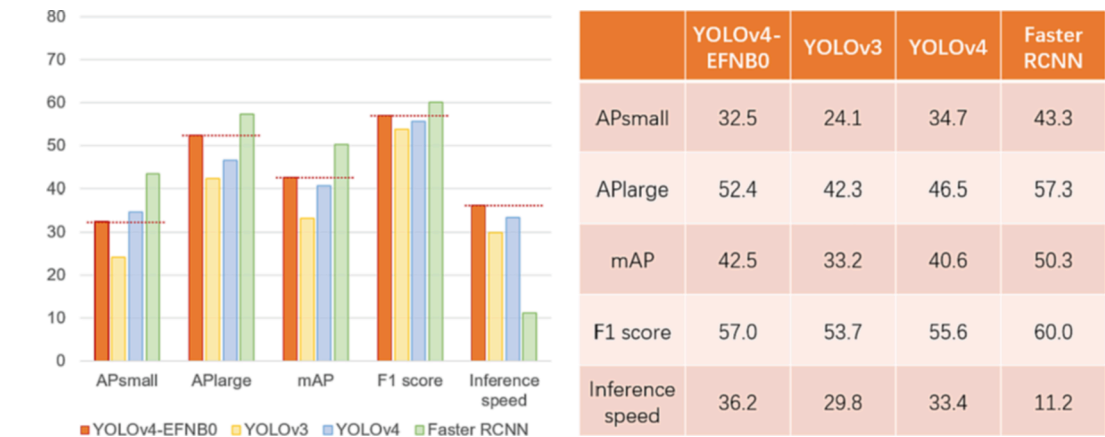


Fig. 14. Comparison of the state-of-the-art models trained on MOCS: YOLOv3, YOLOv4, and Faster R-CNN with ResNet-152.

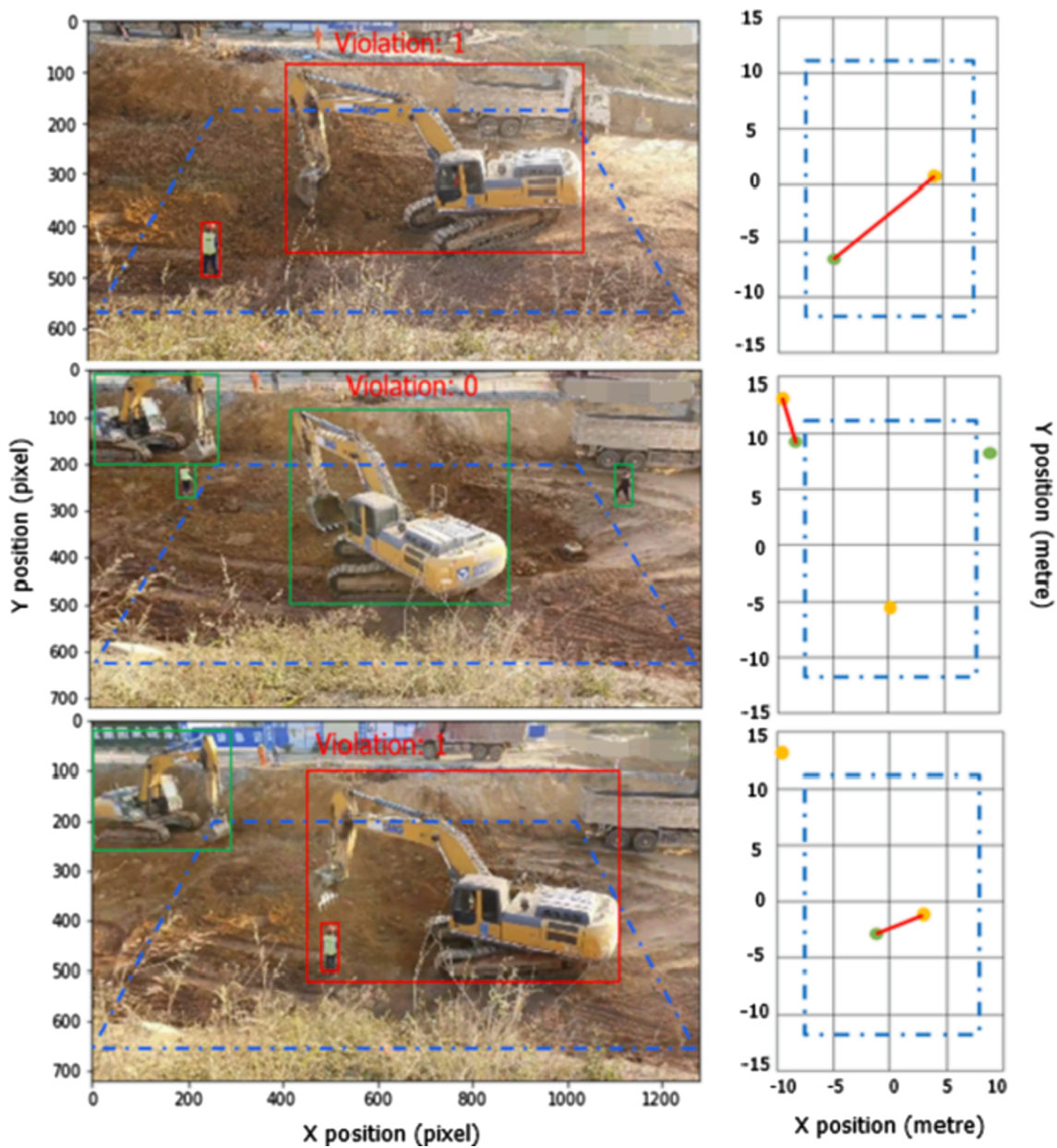


Fig. 15. Demonstration of proximity detection using YOLOv4-EFNB0 and homography transformation.

excavator, and d_g is the ground truth of the distance. The MAE of the violation ratio was also selected as the performance matrix:

$$MAE_r = \frac{1}{n} \sum_{i=1}^n \left| \frac{v_i}{n_i} - \frac{v_g}{n_g} \right| \quad (17)$$

where n is the number of workers inside the projective view and v is the number of workers who violate proximity inside the projective view. There are two methods to calculate the distance between two bounding boxes. The BB-centroid method measures the distance between the centroids of two bounding boxes. The BB-bottom method measures the distance between the bottom centre points of the bounding boxes. The results of the MAE_d and the MAE_r are shown in Table 5. It is noted that the BB-bottom approach is relatively better than the BB-centroid approach. The main reason for this is because of the camera angle. If the camera view is perpendicular to the field, using the BB-centroid approach is more accurate, and vice versa.

Due to the better performance of the BB-bottom approach, we selected it to evaluate the accuracy rate (Eq. (18)) of violation detection in the test video. Each video frame's confusion matrix of violation detection was developed (Table 6). It can be seen that the accuracy can reach 96.76 %. Due to the light-weight characteristic of YOLOv4-EFNB0, the average processing speed is approximately 25 FPS, which achieves near real-time performance.

$$Accuracy = \frac{TP + TN}{TP + FP + FN + TN} \quad (18)$$

There is still some missed detection during testing. As shown in Fig. 16 at frame #276, a worker behind the excavator was not detected temporarily. The main reason for this is that the excavator heavily occluded the worker. Since the application was run on video footage, the occurrence of missed detection in some frames is acceptable. The proposed method is still beneficial for construction safety monitoring, although it misses some detection. This application performs robustly with normal occlusion at frames #36, #186 and #336.

5.4. Practical implications

Existing DL models cannot sustain detection and tracking when workers are partially occluded. Proximity detection would be inaccurate if construction entities cannot be consistently tracked, increasing the accident occurrence probability. Therefore, enhancing DL model performance in occlusion situations facilitates more efficient site safety monitoring. The proposed system allows the proximity status of each worker and construction equipment to be monitored constantly and correctly, even in a cluttered construction site.

Several practical considerations related to the implementation of the proposed system are examined, including the impact on construction site operations, computing efficiency, and setup costs. First, it is practical and feasible to adopt the system to construction sites since installing several surveillance cameras in construction sites is a regular practice (Yang et al., 2018). Cameras should be positioned at a certain height as it optimises the vision field and assures the homography transformation accuracy. Several studies have proved the feasibility of deploying far-field cameras to monitor the movements and behaviours of construction site employees (Assadzadeh et al., 2021; Fang et al., 2018b).

Second, the monitoring system offers excellent computing efficiency in video processing, making it possible to send alerts to safety inspectors

Table 5
Proximity detection performance.

Approach	MAE_d	MAE_r
BB-centroid	1.787 m	0.074
BB-bottom	0.684 m	0.036

Table 6
Confusion matrix of proximity violation.

		Ground truth		Total
		Violation	No violation	
Predicted	Violation	1678	84	1762
	No Violation	59	2589	2648
	Total	1737	2673	4410
	Accuracy	0.9676		

in real-time. The light-weight DL model with GPU batch processing achieves fast inference through parallel computing, even when several workers and construction equipment are in a single frame. Overall, the system can perform video inference at a rate of around 25 FPS, which allows proximity hazards to be proactively detected.

Last but not least, implementing the system in construction sites would be economical. The size of the developed DL model has been significantly reduced while performance has barely decreased, which is suitable to be integrated with surveillance cameras. Studies have recommended several methodologies for the layout of camera arrangements, which would optimise the field of view captured by each camera (Albahri and Hammad, 2016; Heyns, 2021). Hence, the hardware and software setup is deemed cost-effective. With the implementation of this low-cost automation process, the construction industry may devote more resources to safety education/training, planning, and inspection.

6. Conclusion and future work

Although DL applications have evolved recently, their use in the construction sector is still in an early stage. Due to the frequent occurrence of collision incidents in construction sites, DL-based proximity detection has a great potential to improve construction safety. It is likely that certain fundamental challenges still remain, and one of these is perceived as recognition inaccuracy due to occlusion problems in cluttered scenes. Another challenge is that existing proximity detection applications are impractical as they usually demand high computation power. To effectively improve construction site safety, it is essential to address these challenges. This study optimises current proximity monitoring methods through context-guided data augmentation, CNN weight reduction, and homography transformation. This study provides new insights into the optimisation strategy of vision-based proximity detection in terms of operational speed and accuracy enhancement, which can substantially promote safety monitoring automation. The significance and contribution of this work are as follows:

- A novel data augmentation method was proposed. This method can overcome DL training data shortage by leveraging image contextual information without compromising algorithmic accuracy. The method augments images by stitching occlusion patterns, which provides DL models with more varieties of training samples to adapt to the cluttered construction scenes.
- A light-weight DL model named YOLOv4-EFNB0 was formulated. As compared to YOLOv4, YOLOv4-EFNB0 is significantly downsized. The experimental results show that when the context-guided data augmentation was employed, YOLOv4-EFNB0 outperformed most of the current DL models in terms of mAP, F1, and processing speed. Overall, this model is less costly and will motivate safety practitioners to use DL-based safety monitoring applications.
- A proximity detection application was developed based on YOLOv4-EFNB0 and homography transformation. This application can achieve an outstanding accuracy rate and a promising inference speed through a feasibility test, namely, 96.76 % in 25 FPS. By promptly identifying the close-by hazards, this application manifests significant merits in risk mitigation. In addition,

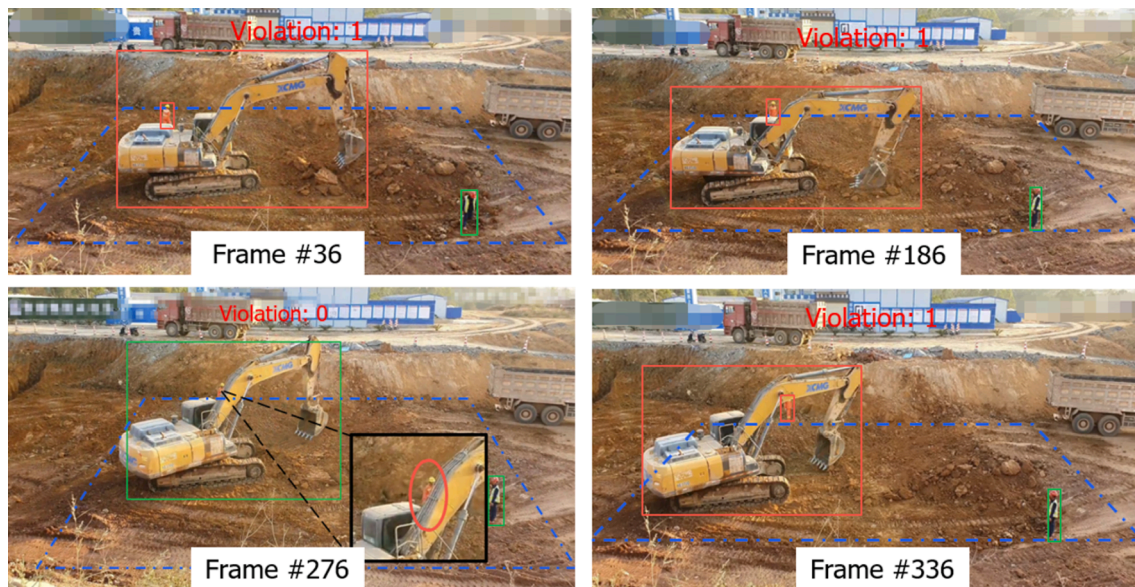


Fig. 16. An example of worker detection failure.

the real-time data collection and processing features boost the automation of construction work sites.

The future work will focus on further dataset augmentation to provide more high-quality training samples and to capture more risk types in occlusion conditions. In addition to providing safety alerts from single camera views, the proposed system can benefit multi-camera systems where the occlusion issues can be resolved to the greatest extent (Elharrouss et al., 2021; Soltani et al., 2018). Besides, the internet of things (Chen et al., 2021), cloud computing, and advanced visualisation concepts will be interfaced with our development to provide a more intelligent, autonomous, and cost-saving safety monitoring solution.

CRedit authorship contribution statement

Haosen Chen: Conceptualization, Data curation, Writing-original draft, Writing-review & editing, Investigation, Validation, Formal analysis, Methodology. **Lei Hou:** Writing – review & editing, Supervision. **Shaoze Wu:** Writing – review & editing, Validation, Methodology, Investigation, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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